can use another program to make an HTTP connection. The text-only browser, Lynx is well suited for this purpose.

Remember, the Web server executes PHP scripts. Executing scripts from the root user's crontab will allow them greater ability to do damage. It is probably best to execute the script from the Web server's crontab. Using Lynx to run the script avoids this issue but raises another. Unless you put the script in a protected directory, anyone will be able to run it. Simply protect the script with a username and password. Lynx will allow you to specify these on the command line.

9.9 EFFICIENCY AND DEBUGGING

Efficiency should not be your first concern when writing code. You must first write code that works, and hopefully your second concern is keeping the code maintainable.

You will pick up some tactical design issues as you gain more experience in programming. These begin to gel as idioms—repeated structures applied to similar problems. Individuals and organizations tend to develop their own idioms, and you will notice them in code found in magazine articles and code repositories. Once you accept an idiom as your own, you can consider it a solved problem.

In most projects, a tiny minority of code is responsible for most of the execution time. Consequently, it pays to measure first and optimize the slowest section. If performance increases to acceptable levels, stop optimizing.

When a bug appears in your script, the time you spent writing meaningful comments and indenting will pay off. Sometimes just reading over troublesome code will reveal its flaws. Most of the time you will print incremental values of variables to understand the problem.

9.9.1 Measuring Performance

Three factors affect the time it takes to go from clicking on a link to seeing a completed Web page. First is the network. Your request must travel from the browser to the server, and the Web page must travel back to your browser.

The best way to measure how long a script runs is to print the time in important points in your script. Because most scripts take less than a second to run, you must use the microtime function. If you place the output in HTML comments, the display of the page will not be disturbed. Of course, the print statement itself will take some time. You can minimize this by simply printing the output of microtime instead of trying to convert its output into an integer. You can do the math later by hand or in a spreadsheet. The first HTML comment contains the time on the clock when the script begins. That's followed by time when the 10,000 cosine calculations have finished. Finally we see the time when the 10,000 lines are written to a file.